KAYD SOLUTIONS

Independent Economic & Technical Evaluation

Engagement Context Pack

Full handoff record — for continuation in a desktop session

MWSC × BECO — proposed water-utility merger / joint venture. Everything needed to resume the engagement with full context: mandate, transaction, ownership question, standards, methodology, delivered work, queued workstreams, open questions, and a ready-to-paste kickoff prompt for desktop.

Prepared by	Kayd Solutions (independent evaluator)
Repo / branch	idbac25/Bayzara · claude/water-utility-merger-research-Qh7FK
Date	2026-06-12
Status	Phase 1 in progress; borehole audit reference delivered
Use	Carry to desktop cloud conversation to continue detailed document work

0. What this document is

This is a **portable context pack** for the MWSC × BECO independent economic evaluation. Its job is to let you (or any Claude session — desktop, web, CLI) **resume this engagement with full context** without re-explaining anything. It compresses the entire conversation to date: the mandate, the transaction, the unresolved ownership question, the applicable standards, the agreed methodology, the work already delivered, what is queued, and the open questions. Section 13 gives a ready-to-paste kickoff prompt for a fresh desktop conversation.

> **PURPOSE OF THE MOVE**
> Why move to desktop: this session runs against a GitHub repo, which is ideal for committing files but awkward for long-form, heavily-annotated document drafting. Carry this pack over to a desktop cloud conversation to continue producing comprehensive, cleanly annotated documents. All work so far is committed in the repo (Section 9).

1. Engagement snapshot

Field	Value
My role	Independent economic evaluator — neutral, loyal to method & standards, NOT an advisor to either party. Symmetric treatment of both sides.
Mandate order	Phase 1 = evaluate the evaluation methods vs international standards. Phase 2 = derive/test financial values. Methods first, numbers second.
Transaction	Proposed merger/JV: BECO (acquirer) !MWSC (target/operator). Fixed control split 51% BECO / 49% MWSC.
Sector	Water utility (Mogadishu); coastal, semi-arid aquifer; MWSC wellfield near Afgooye.
Current workstream	Asset evaluation — boreholes & wells (delivered); reservoirs, pumps, generators queued.
Provider / author	Kayd Solutions (independent evaluator).
Repo / branch	idbac25/Bayzara · branch claude/water-utility-merger-research-Qh7FK
Status	Phase 1 in progress. Borehole audit reference delivered 2026-06-12.

2. My role & mandate

I act as an **independent economic evaluator**, not an advisor to either party. Loyalty is to **method and standards**, applied **symmetrically to both sides**. The mandate has two phases in strict order:

1. **Phase 1 — Technical / methodology evaluation:**
   evaluate the *evaluation methods* against international standards. You do not validate a number produced by a method you have not first validated.
2. **Phase 2 — Financial evaluation:**
   only after the method passes Phase 1, derive and test the actual values, ranges and sensitivities, and report an independent range.

> Guiding principle: **methods first, numbers second.**

3. The transaction

Parties

- **MWSC**
  — Mogadishu ("Moktoshu") Water Supply Company — the target/operator.
- **BECO**
  — Banadir (Banaadir) Electric Company — the acquirer. (User sometimes writes "BECCO"/"Beko".)

BECO — who they are (web-verified)

- Somalia's
  largest electricity utility
  ; founded
  5 May 2014
  by
  merging
  the major Banadir-region electricity companies !ả
  serial consolidator with a repeatable absorption playbook
  .
- Covers ~

80% of Mogadishu plus Jubaland, Southwest, Hirshabelle; distributes to Kismayo, Barawe, Marka, Balad, Jowhar, Afgooye, Elasha.

- Well-capitalized: **$220M green-energy program**, ~30% renewables; solar cut tariff **$0.49 !$0.36/kWh**.
- **A power company expanding sideways into water** (new vertical), NOT a water company.

Deal-structure facts (per user — keep neutral)

- **Equity split is a FIXED control rule:** BECO **51%** / MWSC **49%** — **predetermined, NOT derived from the valuation.** (An earlier "control-lever / you'd own 58%" reasoning was **withdrawn** once this was clarified.)
- The **49%** sits in a **ring-fenced MWSC subsidiary only** — MWSC shareholders do **not** share in BECO's wider group upside.
- BECO applies a **30% downward "haircut" to MWSC's assets only** (asymmetric/one-sided) — basis unknown.
- BECO offers a discretionary **~$3M "goodwill" sweetener.**
- **BECO's own contribution for its 51% is UNKNOWN** (cash? folded-in assets? capital + management?) — open question.

Strategic / contextual reads (neutral)

- Because the split is fixed, the live question is: **"Since the split is fixed, what does the 30% haircut actually adjust?"** Candidates: (a) recorded capital accounts of the sub, (b) fairness basis for the $3M, (c) a future buyout/exit formula for the 49%. If "nothing, just bookkeeping," it has no analytical basis.
- **Related-party risk (neutral disclosure):** BECO would be both **controlling parent AND electricity supplier** to the water sub !transfer-pricing / management-fee channels could move profit from the 49%-shared sub to the 100%-owned parent. Methodology must disclose/handle this.
- **Synergy of record:** BECO already powers Afgooye/Elasha (where MWSC's wellfield sits). Pumping energy (diesel) is a water utility's biggest OPEX; BECO's (solar) grid could slash it.
- BECO's template was built to absorb **small diesel gensets** (commodity, replaceable). MWSC is a **strategic, hard-to-replicate aquifer system** — a neutral methodology should not assume the template fits.

4. Ownership / legal situation (critical scoping fact)

- After the **1991 collapse of central government**, MWSC has for **~35 years** managed, maintained, protected and **added substantial value** to a formerly **public** water system.
- **There is NO formal contract/concession** — only **tacit government approval** to operate.
- Therefore **legal title is unresolved.** The evaluator must NOT assume either extreme ("all public" vs "all MWSC's"). (This walked back an earlier "it's all public / IPSAS governs the lot" framing.)

Neutral valuation treatment agreed

1. **Separate value from ownership.** Value the asset base by DRC / income approach regardless of owner; attribute ownership per a **stated special assumption (IVS 101/102)** taken from a **qualified Somali-law title opinion (IVS 104 reliance on a legal specialist)** — the evaluator does not adjudicate title.
2. **Asset-vintage split:** after 35 years most pre-1991 physical assets are depreciated or replaced; **most in-service assets are likely MWSC-funded.** Date each asset by class; let depreciation reveal the split (evidence-based).
3. **Quantify "value added" in three buckets:** (a) MWSC self-funded capex/betterment (cost approach); (b) going concern & intangibles — customer base, billing/collections, operating capability, brand (**IVS 210** / income approach); (c) potential **improvement / unjust-enrichment compensation claim** if title reverts (subject to legal opinion).
4. **Title-risk discount:** a purely tacit, revocable operating right is fragile !’ reflected **symmetrically** as an income-approach risk input (horizon / discount rate), applied to the whole enterprise regardless of holder.
5. **Present value under multiple title scenarios** so the conclusion is transparent, not hostage to an unresolved legal question.

> **DECISION PENDING**
> Open fork: does a written Somali-law title/legal opinion exist, or can one be commissioned? This decides whether the methodology presents a single title basis or a multi-scenario matrix.

5. Applicable world standards

Valuation

- **IVS 2025** (effective 31 Jan 2025): IVS 101 Scope, 102 Bases of Value, 103 Approaches, 104 Data & Inputs, 105 Models, 106 Documentation; **IVS 200 Businesses**, **IVS 210 Intangibles**.
- **IVS 102 Equitable Value** likely more appropriate than Market Value (transfer between two specific identified parties).
- **IVS 200:** cost approach "rarely applicable" to a cash-generating concession !’ **income approach (DCF) primary**; **DRC (cost approach)** for specialized physical assets with no active market.

- **IPSAS 45 Property, Plant & Equipment** (effective 1 Jan 2025; replaced IPSAS 17): governs public-sector infrastructure (names water supply systems); recognizes **DRC** / current operational value. **Applies only to the genuinely public portion.**
- **RICS Red Book Global** — operationalizes IVS.

Borehole / water engineering (Phase-1 technical)

Full, verified citations are in Section 10. Spine: ISO 14686; Kruseman & de Ridder (ILRI 47); AWWA A100 / M21; WHO GDWQ (2022); ISO 5667 sampling; SANS 10299-4; Kenya WRA test- pumping code; US EPA borehole geophysics; USGS groundwater/seawater-intrusion.

6. Phase-1 methodology-conformance checklist (apply to BOTH parties)

1. **Object of valuation (IVS 101):** carve the estate into *original public asset / MWSC-funded improvements / intangibles & going concern.*
2. **Basis of value (IVS 102):** declare & justify (Equitable Value likely); state title special assumption(s).
3. **Approach selection (IVS 103/105, 200):** is DRC right per asset class? Income approach for the revenue-earning concession?
4. **Consistency & symmetry:** are the same methods/assumptions/adjustments applied to both sides? (The **30% one-sided haircut** is a consistency/objectivity test, not a grievance.)
5. **Data & inputs (IVS 104):** replacement costs, useful lives, depreciation sourced & reliable; reliance on legal title opinion noted.
6. **Documentation (IVS 106):** transparent, reproducible.
7. **Independence/objectivity:** disclose related-party features (BECO controlling + supplying; political ties) as risk factors.

> Phase-1 deliverable: a **methodology-conformance scorecard** rating the approach against IVS 2025 + IPSAS 45 (conforms / partial / non-conforming) with the gap and standard reference for each.

7. Borehole & well evaluation plan

Context: coastal, semi-arid aquifer; MWSC wellfield near Afgooye. Lead value drivers: **(a) sustainable yield vs over-abstraction** and **(b) saltwater intrusion / salinity trend.** A structurally perfect borehole is worth little if water turns brackish or the table is falling. Per-module template: **Authority !Method !Metric !Valuation input.**

Module	Method / authority	Feeds (valuation input)
M1 Structural integrity	CCTV, caliper, gamma/casing logs, verticality, sanitary-seal inspection (AWWA A100; US EPA geophysics)	Remaining structural life !DRC depreciation; rehab-vs-replace cost
M2 Hydraulic performance & yield	Step-drawdown (3+ rates "e60min) !'efficiency; constant-rate (±5%, 24–72 h) !'sustainable yield + T,S; recovery !' cross-check T (ISO 14686; Kruseman & de Ridder; Kenya WRA)	Revenue capacity; ageing indicator (specific-capacity decline)
M3 Aquifer sustainability	Long-term water-level monitoring; recharge/water balance; well-interference (USGS sustainable-yield framework)	Asset life horizon (income approach); over-abstraction risk !'discount-rate driver
M4 Water quality	Lab panel, representative + seasonal (WHO GDWQ 2022; Water Safety Plan). E.coli 0/100mL; As 0.01; F 1.5; NO 50; salinity suite TDS/EC/Cl/Na/SO „	Treatment OPEX; salinity caps price & life
M5 Saltwater intrusion (coastal overlay)	EC depth profiling; chloride trend; resistivity interface mapping; Ghyben–Herzberg (USGS seawater intrusion)	Potentially the single largest impairment

Synthesis !RUL & risk grade per borehole

- **RUL = minimum of**
  structural life (M1), aquifer life (M3), quality/salinity life (M4–M5).
- **Capacity = sustainable yield**
  (M2), de-rated for blending losses.
- **OPEX = pumping energy + treatment**
  (M4) — note BECO power synergy.
- RUL + capacity + OPEX !'income approach; RUL + condition !DRC depreciation.

Tiered execution (many boreholes)

1. **Tier 1 — Census (all):**
   ID, location, depth, construction date, casing/screen spec, pump, metered output, static level + EC spot.
2. **Tier 2 — Rapid field screen (all):**
   wellhead condition, water level, EC/salinity spot, specific-capacity spot check.
3. **Tier 3 — Full assessment (representative + critical subset):**
   CCTV + caliper, full constant-rate test, full WHO lab panel, EC depth profile. Extrapolate to population.

8. What has been delivered

Borehole & Well Evaluation Reference (delivered 2026-06-12) — a standalone authoritative **audit reference** for both non-technical reviewers and technical evaluators. Lens: the user is **auditing the evaluators** (with no instruments), so it states what a competent evaluation must measure, by which method, to which named standard, how it is scored, and the valid **low-/no-cost substitutes** when equipment is missing.

- **Outputs:**
  branded PDF (Kayd Solutions) + unbranded PDF + Markdown source; single-source generator (pdfmake). 17 pages.
- **Contents:**
  how-to-use; plain-language primer; 4 dimensions (Structural / Yield / Quality / Sustainability & saltwater intrusion) each with terms, gold-standard method, no-equipment fallback, governing standard, "good" thresholds, audit checklist, scoring; a ready-to-use scoring rubric (
  BCSI 0–100 + A–E grade + weakest-link RUL ceiling
) with a worked example; the value bridge (RUL/capacity/OPEX !DRC & income approach); auditor red-flags; glossary; verified references.
- **Assurance:**
  all standards & formulas were
  verified by two sub-agents
  before release.

Artifact	Repo path
Branded PDF	docs/mwsc-evaluation/pdf/Borehole-Well-Evaluation-Reference_Kayd-Solutions.pdf
Unbranded PDF	docs/mwsc-evaluation/pdf/Borehole-Well-Evaluation-Reference.pdf
Markdown source	docs/mwsc-evaluation/borehole-well-evaluation-reference.md
Generator (single-source)	docs/mwsc-evaluation/build/generate.js (rebuild: npm i pdfmake@0.3.11 && node generate.js ..)
This context pack	docs/mwsc-evaluation/pdf/ + build/handoff.js
Engagement memory	docs/mwsc-evaluation/ENGAGEMENT-MEMORY.md (the master running record)

9. Verified standards & formulas (reference)

Locked-in citations (independently verified — supersede any earlier "edition TBC" notes):

Standard / source	Verified detail
ISO 14686:2003	Hydrometric determinations — Pumping tests for water wells — Considerations and guidelines for design, performance and use. Current; revision ISO/DIS 14686 pending. Supersedes BS 6316:1992.
ISO 5667-11:2009	Water quality — Sampling — Part 11: Guidance on sampling of groundwaters.
ISO 5667-3:2024	Preservation & handling of water samples (current — NOT 2018).
WHO GDWQ	4th ed. incorporating 1st & 2nd addenda (2022). ISBN 9789240045064.
ANSI/AWWA A100-20	Water Wells (2020).
AWWA M21	5th ed. 2025, retitled "Groundwater Operations" (manual, not a consensus standard).
SANS 10299 / -4:2003	Development, maintenance & management of groundwater resources; Part 4 = Test-pumping of water boreholes (SABS).
DWAF/DWS (SA)	South African Water Quality Guidelines Vol.1: Domestic Use, 2nd ed. (1996).
Kenya WRA	Code of Practice for Test Pumping — posted version is a DRAFT; verify final adoption before formal citation.
US EPA	Environmental Geophysics — Borehole Geophysical Methods (caliper, gamma, fluid temp/conductivity, flow).
USGS GWPD 4	Electric-tape water-level measurement.
Misstear, Banks & Clark (2017)	Water Wells and Boreholes, 2nd ed., Wiley-Blackwell — independent text, NOT a BGS publication.
Kruseman & de Ridder	Analysis and Evaluation of Pumping Test Data, ILRI Pub. 47.
Healy & Cook (2002)	Using groundwater levels to estimate recharge (WTF method).

Formulas confirmed: specific capacity Sc = Q/s; Jacob step-drawdown s = BQ + CQ^2 and efficiency BQ/(BQ+CQ^2); Theis; Cooper–Jacob T = 2.30Q/(4< ”Δs), S = 2.25T·t₀/r²; Theis recovery; Ghyben–Herzberg z "40h; WTF recharge R = Sy·9Fƒ²
TDS "0.64×EC ("H0.75 for seawater-intruded).

10. Queued workstreams

1. **Reservoirs / storage tanks**
   (concrete + steel) — "Module 6" on the same template. *User will provide photos.* Concrete: structural condition, cracking, spalling, rebar corrosion, leakage, capacity. Steel: corrosion, coating, plate thickness, cathodic protection.
2. **Pumps & generators**
   — condition, rating, efficiency, remaining life, replacement cost, energy cost (links to BECO power synergy).
3. **Field protocol + scoring sheet**
   — convert the borehole plan into one row per borehole (Tier 1–3 data fields, auto-RUL/risk grade).
4. **Phase-1 conformance toolkit**
   — "Methodology Conformance — IVS 2025 / IPSAS 45" scorecard + scoping framework (asset-vintage split, value-added buckets, title-scenario matrix).

11. Open questions to confirm (client / "Mahad")

1. **What does BECO contribute for its 51%?**
   (cash / assets / capital + management)
2. **Since the split is fixed at 49/51, what does the 30% haircut adjust?**
   (capital accounts / $3M basis / buyout formula)
3. **What electricity tariff & related-party terms**
   will BECO charge the water subsidiary?
4. **Does a written Somali-law title/legal opinion exist**
   (or can one be commissioned)?
5. Confirm any remaining edition/scope points before final report (most now verified in Section 9).

12. Working / admin notes

- Repo:
  idbac25/Bayzara
  ; branch:
  claude/water-utility-merger-research-Qh7FK
  . Repo is otherwise a Next.js app ("Bayzara"); this is a research/advisory engagement living under docs/mwsc-evaluation/.
- Units to standardize:
  m, m³/day, L/s, mg/L
  . Currency: USD (e.g. $3M) as stated by client.
- PDF tooling: environment had no working Chrome/pandoc/LibreOffice !PDFs built with
  pdfmake
  (pure JS, standard-14 fonts). On desktop you have richer formatting/annotation options.
- Master running record =
  ENGAGEMENT-MEMORY.md
  (keep it updated as the single source of truth).

13. How to continue on desktop — ready-to-paste kickoff

Start a new desktop conversation and paste the block below (attach this PDF too). It re-establishes the mandate and points to the next task.

```
You are continuing an in-progress engagement. Adopt this context fully.

ROLE: I am an INDEPENDENT economic evaluator (Kayd Solutions) for a proposed
merger between BECO (Banadir Electric Company, acquirer, 51%) and MWSC
(Mogadishu Water Supply Company, target, 49%). You are neutral, loyal to method
and standards, and treat both parties symmetrically. Mandate: PHASE 1 evaluate
the evaluation METHODS vs international standards, THEN Phase 2 derive financial
values. Methods first, numbers second.

STANDARDS SPINE: Valuation = IVS 2025 (esp. 101/102/103/104/200/210; Equitable
Value) + IPSAS 45 (public portion only). Engineering = ISO 14686:2003, AWWA
A100-20 / M21 (2025), WHO GDWQ 2022, ISO 5667-11:2009 & -3:2024, SANS 10299-4,
Kenya WRA test-pumping (draft), US EPA geophysics, USGS.

KEY FACTS: ownership/title is UNRESOLVED (no concession, only tacit approval,
~35 yrs MWSC operation) - never assume "all public" or "all MWSC". Split is FIXED
51/49 (not derived from valuation). 30% haircut applies to MWSC assets only -
basis unknown. ~$3M goodwill sweetener. Coastal semi-arid aquifer near Afgooye;
lead risks = sustainable yield + saltwater intrusion.

DONE: a 17-page Borehole & Well Evaluation audit reference (attached pack has
full detail). QUEUED next: reservoirs/tanks (concrete + steel) "Module 6" - I will
send photos; then pumps & generators; then the field scoring sheet; then the
Phase-1 IVS/IPSAS conformance scorecard.

TASK NOW: [state what you want - e.g. "build the reservoirs/tanks module from
these photos" or "draft the Phase-1 conformance scorecard"].
```

> Everything referenced here is committed in the repo under docs/mwsc-evaluation/. If you want the desktop session to see the actual files, upload this pack plus ENGAGEMENT-MEMORY.md and the borehole reference PDF.

End of context pack.